Predicting Success of Newly Hired Power-4 Head Football Coaches

A Regression, Model-Selection and Survival Analysis

Project Report
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Abstract

Universities spend large sums hiring head football coaches, yet it is unclear which information available *before* the hire actually predicts how well the coach will perform. Using a dataset of [132] coach hires and [60] variables, we model the coach's impact on team strength (`sp_rel_team`) using linear regression, and we model retention and tenure using logistic regression and survival analysis. We test the regression assumptions, treat heteroscedasticity and multicollinearity, select a parsimonious model with information criteria and penalised methods, check its stability using the bootstrap and jackknife, account for school-level clustering with mixed models, and validate the model on a prospective holdout of hires from 2022 onward. The best model reaches a cross-validated R^2 of [**0.0708**] and a prediction-interval coverage of [**0.952(or 95.2 %)**] on the holdout. The main predictors were PreviousOC (previous offensive coordinator experience) and team_sp_recent (recent team SP+ performance context). We conclude with a simple "Coach Hire Score" decision rule.

1 Introduction and Objectives

A head coach is one of the most visible and costly hires a Power-4 football programme makes. Athletic directors usually rely on reputation and intuition. This project asks whether a statistical model built only on *pre-hire* information can predict the coach's later success.

1.1 Research questions

Q1. Which pre-hire characteristics (prior head-coaching record, coordinator background, NFL experience, age, ties to the school, recent team strength) are associated with a coach's impact on the team?

Q2. Do the standard linear-model assumptions hold, and how should violations be handled?

Q3. Is the selected model stable under resampling, and does it predict genuinely new hires?

Q4. Does clustering of hires within schools or conferences matter?

Q5. Do the same factors predict whether a coach is retained for four seasons and how long his tenure lasts?

1.2 Outcomes studied

- **Primary outcome:** `sp_rel_team`, the coach-era SP+ rating relative to the team's level before the hire (referred to as CPI in our scripts). Positive values mean the team improved relative to its previous level.
- **Retention outcome:** `Retained4`, a binary indicator equal to 1 if the coach completed at least four seasons without leaving for off-field reasons.
- **Tenure outcome:** number of seasons until the coach is fired (right-censored otherwise).

1.3 Report structure

Section 2 describes the data and preparation. Section 3 gives the statistical theory behind every method used. Section 4 presents the results. Section 5 describes the decision rule. Sections 6 and 7 discuss limitations and conclude.

2 Data

2.1 Source and structure

The raw file (`data/raw/data.csv`) has one row per coach hire, with [132] rows and [60] columns. The variables fall into the groups in Table 1.

Table 1: Groups of variables in the dataset.

Group	Variables (examples)
Identity	`name`, `school`, `first_season`, `last_season`, `Conf.Hired`
Prior career	`PrevHCAnyLevel`, `PreviousOC`, `PreviousDC`, `SideOfBall`, `NFLCoachingExperience`, `numb_years_Coord_HC`
Prior record	`PreviousWinPct`, `AllPriorWinPct`, `Wins.at.Prev`, `Losses.at.Prev`
Achievements	`WonCCasHCany`, `WonNCasHCany`, `WonCCasCoordAny`
Ties	`ConnectionToHiring.School`, `Regional.Ties`, `PromoteWithin`, `SameConf`
Personal	`Birth.Year`, `AgeAtHiringMinus50`, `PlayedNFL`, `Played.Power.4`
Outcomes	`Fired`, `OffFieldIssuesFiring`, `StillEmployedAtSchool`, `WinsPostHire`
Performance	`coach_sp`, `sp_rel_team`, `winpct_rel_team`, `srs_rel_team`
Team context	`team_sp_prev`, `team_sp_recent`, `sp_prev_rel`

Many numeric columns are stored as text in the raw file. A helper function (`auto_num`) converts a character column to numeric whenever every non-missing entry is numeric.

2.2 Sample restriction

To measure a coach's effect we need enough seasons of data. The analysis sample (`reg_df`) keeps coaches who

- completed at least two seasons (`seasons_completed` = `last_season` − `first_season` +1 ≥ 2), and
- either left the school or were hired in 2021 or earlier (`StillEmployedAtSchool`= 0 or `first_season`≤ 2021).

This leaves [84] observations. One coach (Sam Pittman, Arkansas) was excluded because he was flagged as a prior head coach but had no recorded prior wins or losses, so his prior win percentage was a placeholder rather than real data. Imputing from it would have been circular.

2.3 Handling missing values

- `AllPriorWinPct` and `PreviousWinPct` are missing by construction for first-time head coaches (`PrevHCAnyLevel`= 0). They are set to a neutral 0.50.
- `sp_prev_rel` is missing for coaches without a previous head job. It is filled with the sample mean. In the model it enters only through the interaction `PrevHCAnyLevel:sp_prev_rel`, so the filled value is multiplied by zero for first-time head coaches.
- [2] prior head coaches had missing `AllPriorWinPct` and were filled from `PreviousWinPct`; these rows are logged in `tables/allpriorwinpct_imputation_log.csv`.

2.4 Data-quality flags

Five coaches had inconsistent employment fields (Table 2). Two were kept and flagged. Three were excluded in a sensitivity analysis of the survival model.

Table 2: Flagged records (from `data_corrections.csv`).

Coach (school)	Issue	Action
Hugh Freeze (Auburn)	`Fired`=1 and `StillEmployed`=1	Retained, flagged
Luke Fickell (Wisconsin)	`Fired`=1 and `StillEmployed`=1	Retained, flagged
Brian Kelly (LSU)	External record disagrees	Excluded in sensitivity
Mike Gundy (Oklahoma State)	External record disagrees	Excluded in sensitivity
James Franklin (Penn State)	External record disagrees	Excluded in sensitivity

2.5 Descriptive statistics

Table 3: Descriptive statistics of key numeric variables (`descriptive_stats.csv`).

Variable	Mean	SD	Median
`sp_rel_team`	[-1.96]	[8.44]	[-2.15]
`winpct_rel_team`	[-0.02]	[0.15]	[-0.01]
`AllPriorWinPct`	[0.64]	[0.17]	[0.66]
`PreviousWinPct`	[0.58]	[0.12]	[0.50]
`AgeAtHiringMinus50`	[-3.89]	[6.86]	[-5.00]
`team_sp_recent`	[7.83]	[8.91]	[9.42]

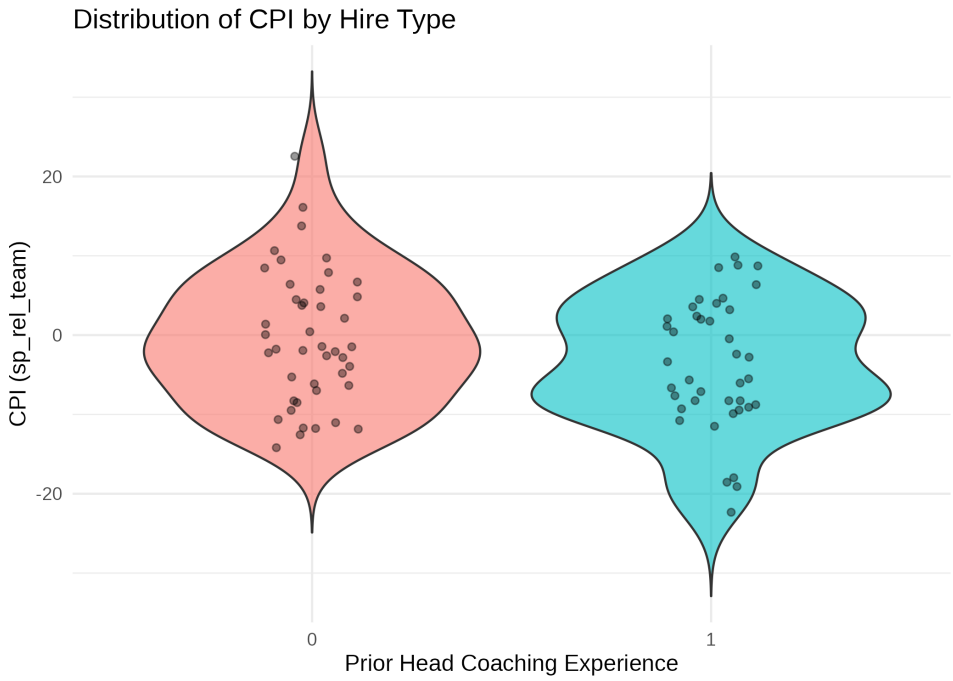


Figure 1: Distribution of the outcome `sp\_rel\_team` by type of hire.

2.6 The full model formula

All regression models use the following full ("confirmatory") specification:

$$\texttt{sp_rel_team} \sim \texttt{PrevHCAnyLevel} + \texttt{AllPriorWinPct} + \texttt{PreviousWinPct} + \texttt{PreviousOC} + \texttt{PreviousDC} + \texttt{SideOfBall} + \texttt{ConnectionToHiring.School} + \texttt{PlayedNFL} + \texttt{AgeAtHiringMinus50} + \texttt{NFLCoachingExperience} + \texttt{PrevHCAnyLevel:sp_prev_rel} + \texttt{WonCCasHCany} + \texttt{Conf.LeavingP4} + \texttt{team_sp_recent} + \texttt{PromoteWithin}.$$

3 Statistical Theory and Methods

This section explains, in order, each technique used in the project.

3.1 Multiple linear regression (OLS)

For n coaches and p coefficients (including the intercept) the model is

$$y = X\beta + \varepsilon, \qquad \varepsilon \sim (0, \sigma^2 I_n),$$

where y is `sp_rel_team` and X is the $n \times p$ design matrix. The ordinary least squares (OLS) estimator minimises the residual sum of squares and equals

$$\hat{\beta} = (X^\top X)^{-1} X^\top y, \qquad \mathrm{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}.$$

To test $H_0 : \beta_j = 0$ we use $t_j = \hat{\beta}_j / \mathrm{se}(\hat{\beta}_j)$ with $n - p$ degrees of freedom. The *partial correlation* of a predictor with the response, given the others, is recovered from its t value by

$$r_j = \frac{t_j}{\sqrt{t_j^2 + (n - p)}}.$$

Goodness of fit is summarised by $R^2 = 1 - \mathrm{RSS}/\mathrm{SST}$ and by adjusted R^2, which penalises the number of predictors.

3.2 Regression diagnostics

Inference from OLS requires linearity, constant error variance, independent errors and (for small samples) normal errors. We check each.

Residual plots. Residuals against fitted values reveal non-linearity. The normal Q–Q plot assesses normality. The scale–location plot shows whether the spread of residuals changes with the fitted value.

Leverage and influence. The hat value h_{ii} measures how unusual the predictors of observation i are. Cook's distance measures how much all fitted values change when i is removed:

$$D_i = \frac{e_i^2}{p\,\hat{\sigma}^2} \cdot \frac{h_{ii}}{(1 - h_{ii})^2}.$$

DFBETAS measures the change in each coefficient. We flag points using the usual rules of thumb: $D_i > 4/n$, $h_{ii} > 2p/n$ and $|\mathrm{DFBETAS}| > 2/\sqrt{n}$.

Breusch–Pagan test. The squared residuals are regressed on the predictors. Under homoscedasticity the test statistic nR^2_{aux} is approximately χ^2 with degrees of freedom equal to the number of predictors.

White's test. A more general version that also includes squared predictors in the auxiliary regression. Because the sample is small we use squares only and omit cross-products, to avoid using up all the degrees of freedom. The statistic is $nR^2_{\mathrm{aux}} \sim \chi^2_q$ with q the number of auxiliary regressors.

Autocorrelation. Coaches are ordered in time, so errors might be correlated. The Durbin–Watson statistic is

$$DW = \frac{\sum_{t=2}^{n} (e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2},$$

which is near 2 when there is no first-order autocorrelation. We also inspect the ACF/PACF and run a runs test.

3.3 Remedies for assumption violations

Box–Cox transformation. The Box–Cox family transforms a positive response by

$$y^{(\lambda)} = \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda}, & \lambda \neq 0, \\ \ln y, & \lambda = 0, \end{cases}$$

and chooses λ by maximum likelihood. Since `sp_rel_team` takes negative values, we shift it by a constant to make it positive before estimating λ. A value of $\hat{\lambda}$ near 1 indicates that no transformation is needed.

Weighted least squares (WLS). If $\mathrm{Var}(\varepsilon_i) = \sigma_i^2$ varies across observations, the efficient estimator minimises $\sum_i w_i e_i^2$ with $w_i \propto 1/\sigma_i^2$:

$$\hat{\beta}_{\mathrm{WLS}} = (X^{\top} W X)^{-1} X^{\top} W y.$$

We estimate σ_i^2 by regressing the squared OLS residuals on `team_sp_recent` and `AllPriorWinPct`, and set $w_i = 1/\hat{\sigma}_i^2$.

3.4 Multicollinearity

When predictors are strongly correlated, $(X^{\top}X)$ is nearly singular and coefficient variances are inflated. Here `AllPriorWinPct` and `PreviousWinPct` are the obvious candidates, since for many coaches they are nearly the same quantity.

Variance inflation factor.

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 comes from regressing predictor j on all other predictors. Values above 5 are moderate and above 10 are severe. For factors with several levels we use the generalised VIF.

Condition number. For the column-scaled design matrix, $\kappa = \sqrt{\lambda_{\max}/\lambda_{\min}}$ of $X^{\top}X$. Large values (conventionally above 30) indicate near-linear dependence.

Remedies.

- *Iterative VIF pruning:* repeatedly drop the main effect with the largest VIF until all are below 5.
- *Ridge regression:* adds an L_2 penalty,

  $$\hat{\beta}_{\mathrm{ridge}} = \arg\min_{\beta} \|y - X\beta\|^2 + \lambda \|\beta\|_2^2 = (X^{\top}X + \lambda I)^{-1} X^{\top} y,$$

  which shrinks coefficients and stabilises them under collinearity.
- *Principal-components regression (PCR):* regress y on the first k principal components of the standardised predictors, with k chosen by cross-validation.

We compare OLS, ridge and PCR by 10-fold cross-validated RMSE.

3.5 Model building and testing groups of variables

Block F-tests. To test whether a group of q predictors matters jointly we compare a full and reduced model:

$$F = \frac{(\mathrm{RSS}_{\mathrm{red}} - \mathrm{RSS}_{\mathrm{full}})/q}{\mathrm{RSS}_{\mathrm{full}}/(n-p)} \sim F_{q,\, n-p}.$$

We form four blocks: prior head-coach record, coordinator background, ties and pedigree, and hiring context.

Curvature and interaction checks. For each continuous predictor we add a centred squared term and test it with an F-test. Added-variable and component-plus-residual plots show the partial relationship of each predictor with the response. Multiple p-values are adjusted with the Holm method.

3.6 Variable selection and model comparison

Information criteria. For a model with p parameters,

$$\mathrm{AIC} = -2\ell + 2p, \qquad \mathrm{BIC} = -2\ell + p\ln n,$$

where ℓ is the maximised log-likelihood. BIC penalises complexity more heavily and so selects smaller models. Mallows' $C_p = \mathrm{RSS}_p/\hat{\sigma}^2_{\text{full}} - (n-2p)$ should be close to p for a good model, and $\mathrm{FPE} = \frac{\mathrm{RSS}}{n}\cdot\frac{n+p}{n-p}$ estimates prediction error.

Search strategies. We use exhaustive best-subsets search (adjusted R^2, C_p, BIC) and bidirectional stepwise search using AIC and BIC.

Out-of-sample criteria. The PRESS statistic is the leave-one-out prediction error,

$$\mathrm{PRESS} = \sum_{i=1}^{n}\left(\frac{e_i}{1-h_{ii}}\right)^2, \qquad R^2_{\text{pred}} = 1 - \frac{\mathrm{PRESS}}{\mathrm{SST}}.$$

We also use 10-fold cross-validation, reporting CV RMSE and CV R^2.

LASSO and elastic net. The LASSO uses an L_1 penalty,

$$\hat{\beta}_{\text{lasso}} = \arg\min_{\beta} \tfrac{1}{2n}\|y - X\beta\|^2 + \lambda\|\beta\|_1,$$

which sets some coefficients exactly to zero and therefore performs selection. The elastic net mixes L_1 and L_2 penalties. We report the λ with minimum CV error (`lambda.min`) and the sparser λ within one standard error of it (`lambda.1se`).

Bootstrap selection stability and optimism. Selecting a model and then judging it on the same data is over-optimistic. We draw $B = 300$ bootstrap samples, rerun BIC stepwise selection on each, and record (i) the proportion of samples in which each term is selected and (ii) the optimism, i.e. the average difference between the RMSE on the original data and the RMSE on the bootstrap sample. Terms selected in most resamples are considered stable.

Final models. We carry two models forward: the **BIC-selected** model (parsimonious) and the **confirmatory** full model (all pre-specified terms retained, so hypotheses about `AllPriorWinPct` can be tested directly).

3.7 Clustering: random-intercept mixed models

Several coaches were hired by the same school, so their outcomes may be correlated. The random-intercept model is

$$y_{ij} = x_{ij}^{\top}\beta + u_j + \varepsilon_{ij}, \quad u_j \sim N(0,\sigma_u^2),\ \varepsilon_{ij} \sim N(0,\sigma^2),$$

for coach i in school j. The intraclass correlation

$$\mathrm{ICC} = \frac{\sigma_u^2}{\sigma_u^2 + \sigma^2}$$

is the share of residual variance attributable to the school. To test $H_0 : \sigma_u^2 = 0$ we compare the mixed model with the ordinary linear model (both fitted by maximum likelihood) using the likelihood-ratio statistic $2(\ell_{\text{lme}} - \ell_{\text{gls}})$. Because the null value lies on the boundary of the parameter space, the null distribution is a 50:50 mixture of χ^2_0 and χ^2_1, so the correct p-value is half the naive χ^2_1 value. If the optimiser returns a slightly negative statistic, it is floored at zero. Predicted school effects are the best linear unbiased predictors (BLUPs).

3.8 Simultaneous inference

Looking at m coefficients at once inflates the family-wise error rate. With individual level α the confidence intervals are $\hat{\beta}_j \pm c\ \mathrm{se}(\hat{\beta}_j)$ where

$$c_{\text{pointwise}} = t_{n-p,\,1-\alpha/2}, \quad c_{\text{Bonf}} = t_{n-p,\,1-\alpha/(2m)}, \quad c_{\text{Scheffé}} = \sqrt{m\,F_{m,\,n-p,\,1-\alpha}}.$$

Bonferroni is simple and valid for a finite number of intervals; Scheffé is valid for *all* linear combinations and therefore widest. For the mean response over many covariate patterns the Working–Hotelling band uses the same multiplier as Scheffé. We also adjust p-values using Holm (controls family-wise error) and Benjamini–Hochberg (controls false-discovery rate), and compare mean outcomes across conferences with Tukey's HSD and Holm-adjusted pairwise t-tests.

3.9 Logistic regression for retention

For the binary outcome $Y_i =$ `Retained4` we model

$$\ln \frac{\Pr(Y_i = 1)}{1 - \Pr(Y_i = 1)} = x_i^\top \beta,$$

estimated by maximum likelihood. Then e^{β_j} is the odds ratio for a one-unit increase in x_j. We use:

- **Likelihood-ratio tests** for the whole model and for each term (`drop1`).
- **Pseudo-R^2:** McFadden $1 - \ell_1/\ell_0$ and Nagelkerke (a rescaled Cox–Snell measure).
- **Hosmer–Lemeshow test:** groups observations by fitted probability (deciles) and compares observed with expected events; a large p-value indicates adequate fit.
- **ROC and AUC:** the area under the curve of true-positive rate against false-positive rate, i.e. the probability that a random retained coach receives a higher score than a random non-retained coach. The Youden threshold maximises TPR − FPR.
- **Cross-validation:** 10-fold misclassification rate and AUC, compared with the majority-class baseline.
- **Calibration and diagnostics:** observed versus predicted rate by decile, deviance residuals and Cook's distance.
- **Selection and penalisation:** BIC stepwise and LASSO logistic regression, which also guard against (quasi-)separation.

3.10 Resampling: jackknife and bootstrap

Jackknife. Leave out one observation at a time to obtain $\hat{\theta}_{(i)}$ and compute

$$\widehat{\text{bias}} = (n-1)(\bar{\theta}_{(\cdot)} - \hat{\theta}), \qquad \widehat{\text{se}} = \sqrt{\tfrac{n-1}{n} \sum_i (\hat{\theta}_{(i)} - \bar{\theta}_{(\cdot)})^2}.$$

Bootstrap. Resample coaches with replacement $R = 1000$ times, refit the model and collect $\hat{\theta}^*$. The percentile interval uses the 2.5% and 97.5% quantiles of $\hat{\theta}^*$. We apply both to the coefficient of `AllPriorWinPct` and compare with the usual t-interval, to check whether the classical inference is trustworthy for a small, possibly non-normal sample.

3.11 Survival analysis for tenure

Let T be the tenure in seasons. Coaches still employed, or who left for off-field reasons, are *right-censored*; the event is a performance-related firing. The **Kaplan–Meier estimator**

$$\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right)$$

(with d_i events and n_i at risk at time t_i) estimates the survival curve, and the **log-rank test** compares curves between groups (prior head coach versus none). The **Cox proportional-hazards model**

$$h(t \mid x) = h_0(t) \exp(x^\top \beta)$$

gives hazard ratios e^{β_j}. Its key assumption, that hazard ratios are constant over time, is checked with the Schoenfeld-residual test (`cox.zph`).

3.12 Prospective holdout validation

The strongest check uses hires from 2022 onward as a test set. These rows are removed from the training data, both the BIC and the full model are refitted on the training data only, and 95% prediction intervals are computed for the holdout coaches. The *coverage* is the share of true outcomes inside the intervals,

$$\widehat{\text{cov}} = \frac{1}{n_{\text{test}}} \sum_i \mathbf{1}\{y_i \in [\hat{y}_i^{\text{lwr}}, \hat{y}_i^{\text{upr}}]\},$$

which should be near 95% if the model is well calibrated. We give an exact binomial confidence interval for coverage, and the RMSE and MAE of the point predictions. The training-set imputation rules (means and modes) are applied to the holdout to avoid information leakage.

4 Results

All numbers in red are placeholders to be filled from the output tables.

4.1 Baseline model

The full OLS model was fitted on $n =$ [120] coaches, with $R^2 =$ [0.247] and adjusted $R^2 =$ [0.081]. Table 4 shows the coefficients.

Table 4: Full OLS model (`baseline_coefficients.csv`).

Term	Estimate	Std. error	t	p
`(Intercept)`	[0.58]	[8.45]	[0.07]	[0.95]
`PrevHCAnyLevel`	[-1.05]	[3.93]	[-0.27]	[0.79]
`AllPriorWinPct`	[7.72]	[12.33]	[0.63]	[0.53]
`PreviousWinPct`	[-13.49]	[14.09]	[-0.96]	[0.34]
`PreviousOC`	[7.97]	[3.85]	[2.07]	[0.04]
`PreviousDC`	[0.19]	[4.60]	[0.04]	[0.97]
`SideOfBallO`	[-0.51]	[3.15]	[-0.16]	[0.87]
`ConnectionToHiringSchool`	[-0.43]	[2.41]	[-0.18]	[0.86]
`PlayedNFL`	[-3.95]	[3.20]	[-1.23]	[0.22]
`AgeAtHiringMinus50`	[-0.15]	[0.17]	[-0.88]	[0.38]
`NFLCoachingExperience`	[0.80]	[2.24]	[0.35]	[0.72]
`WonCCasHCany`	[4.51]	[2.96]	[1.52]	[0.13]
`Conf.LeavingP4`	[-0.30]	[2.49]	[-0.12]	[0.90]
`team\_sp\_recent`	[-0.27]	[0.12]	[-2.28]	[0.03]
`PromoteWithin`	[3.02]	[3.67]	[0.82]	[0.41]
`PrevHCAnyLevel:sp\_prev\_rel`	[0.08]	[0.17]	[0.48]	[0.63]

4.2 Diagnostics

Table 5: Assumption tests (`heteroscedasticity_tests.csv`, `autocorrelation_test.csv`).

Test	Statistic	df	p-value
Breusch–Pagan	[15.756]	[15]	[0.398]
White (squares only)	[22.56]	[30]	[0.833]
Durbin–Watson	[2.132]	–	[0.736]

Influence. [8] coaches exceeded the Cook's distance threshold $4/n$, [4] had high leverage and [5] had large DFBETAS. The most influential coaches were [**Hugh Freeze, Luke Fickell, and Brian Kelly**].

Interpretation.
[**Breusch-Pagan ($p = 0.398$) and White's test ($p = 0.833$) indicate no significant heteroscedasticity.**]
[**Durbin-Watson ($DW = 2.132$, $p = 0.736$) indicates no first-order autocorrelation.**]
[**Residuals appear approximately normal based on the Q-Q plot.**]

4.3 Remedies

The Box–Cox profile (Figure **??**) gave $\hat{\lambda} =$ [**0.9**] with shift constant [**21**], so [**a transformation was not**] needed. Weighted least squares gave the coefficient comparison in Table 6.

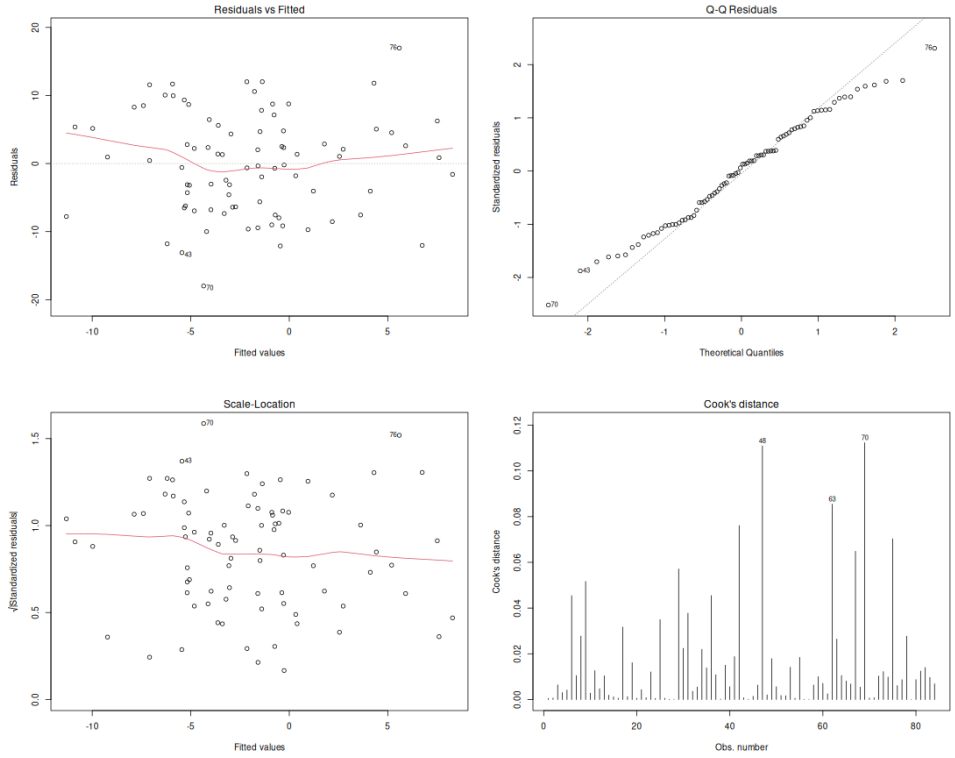


Figure 2: Residuals vs fitted, normal Q–Q, scale–location and Cook's distance for the full model.

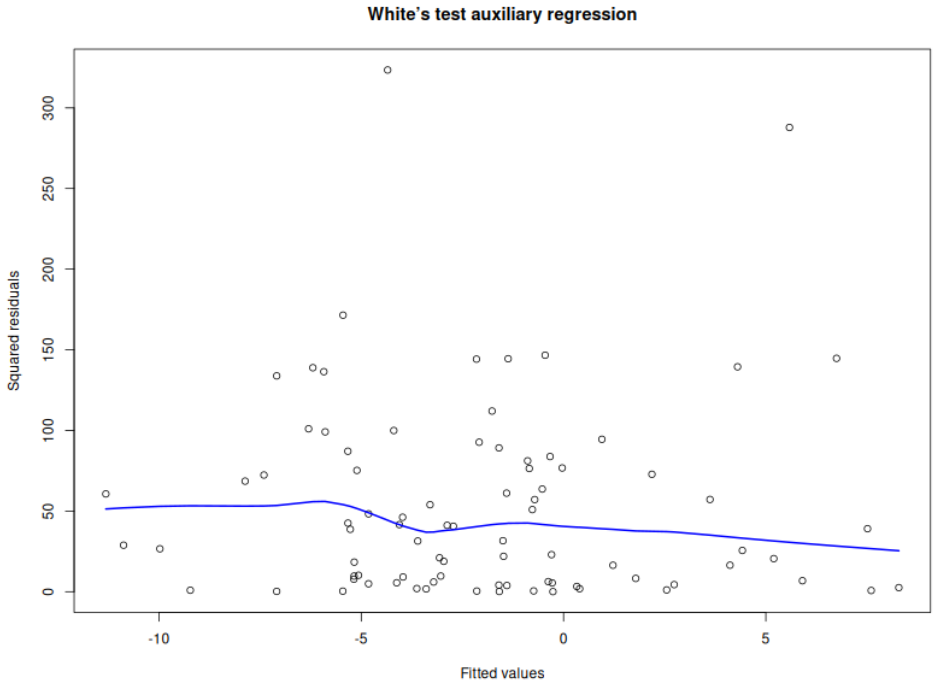


Figure 3: Squared residuals against fitted values with a lowess smoother.

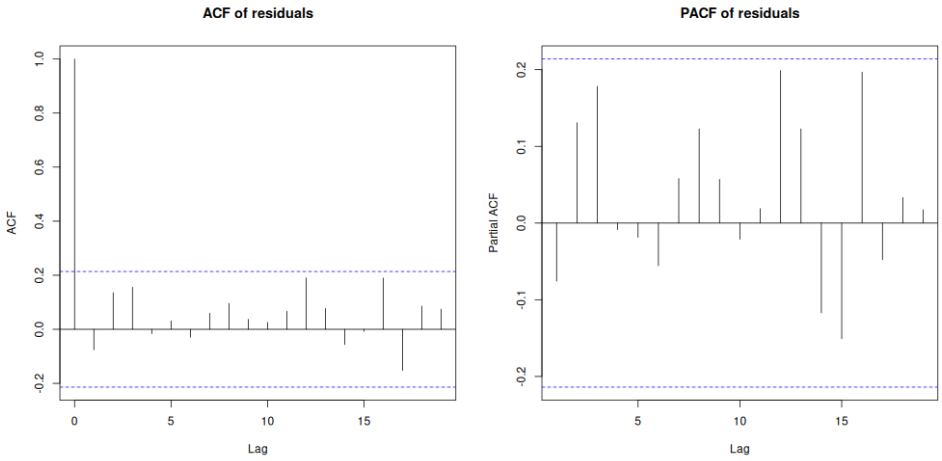


Figure 4: ACF and PACF of the residuals.

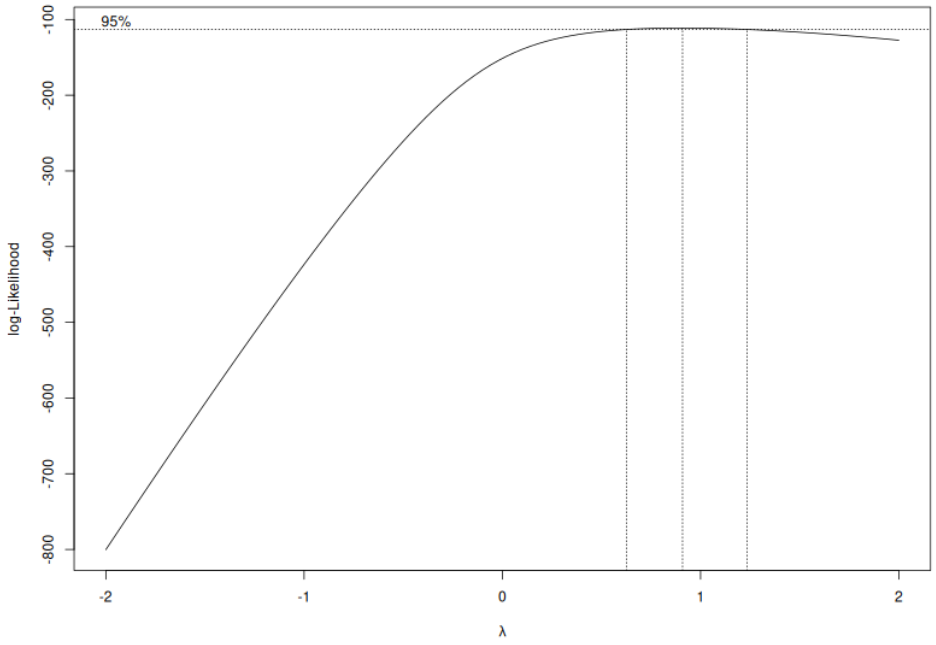


Figure 5: Box–Cox log-likelihood profile for the shifted response.

Table 6: OLS versus WLS coefficients (`wls_coefficients.csv`).

Term	OLS	WLS
(Intercept)	0.58	-1.50
PrevHCAnyLevel	-1.05	-2.28
AllPriorWinPct	7.72	8.83
PreviousWinPct	-13.49	-10.28
PreviousOC	7.97	7.98
PreviousDC	0.19	0.58
SideOfBallO	-0.51	-0.28
ConnectionToHiring.School	-0.43	-0.34
PlayedNFL	-3.95	-3.63
AgeAtHiringMinus50	-0.15	-0.08
NFLCoachingExperience	0.80	1.91
WonCCasHCany	4.51	4.08
Conf.LeavingP4	-0.30	-0.23
team_sp_recent	-0.27	-0.24
PromoteWithin	3.02	2.53
PrevHCAnyLevel:sp_prev_rel	0.08	0.09

4.4 Multicollinearity

The largest variance inflation factors are in Table 7. The condition number of the scaled design matrix is [6.7].

Table 7: Variance inflation factors (`w4_vif_columns.csv`).

Term	VIF	Flag
PreviousDC	[5.55]	[moderate (> 5)]
PrevHCAnyLevel	[4.68]	[]
AllPriorWinPct	[4.11]	[]
PreviousWinPct	[3.59]	[]
SideOfBallO	[3.07]	[]
WonCCasHCany	[2.69]	[]
PromoteWithin	[2.54]	[]
PreviousOC	[2.34]	[]
PrevHCAnyLevel:sp_prev_rel	[1.85]	[]
Conf.LeavingP4	[1.83]	[]
AgeAtHiringMinus50	[1.73]	[]
ConnectionToHiringSchool	[1.71]	[]
team_sp_recent	[1.38]	[]
PlayedNFL	[1.26]	[]
NFLCoachingExperience	[1.25]	[]

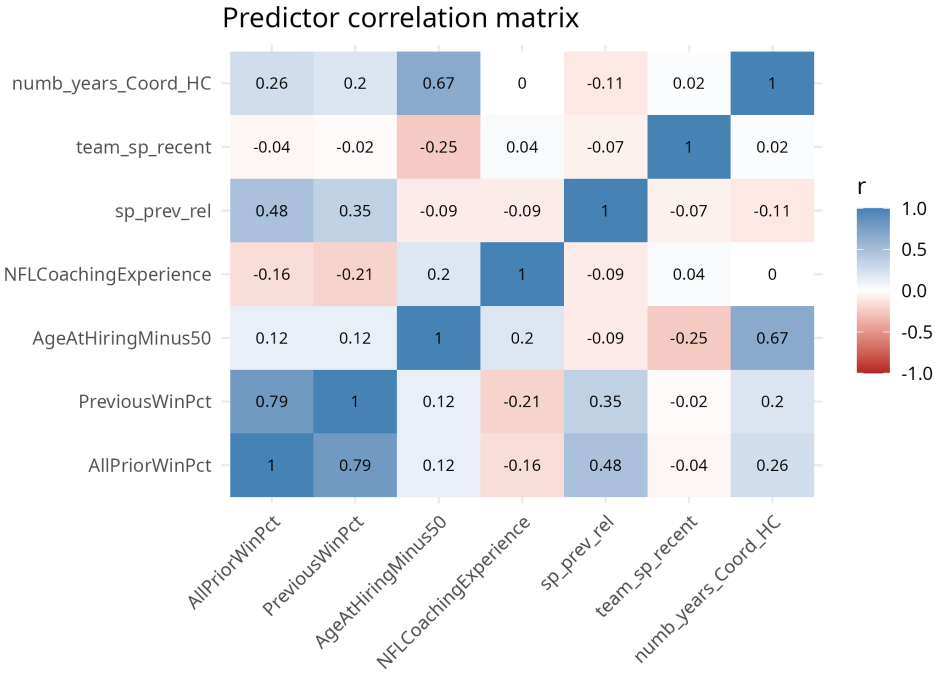


Figure 6: Correlation matrix of the numeric predictors.

Terms dropped by VIF pruning: [**Previous DC**]. The F-test of the pruned model against the full model gave $p =$ [**0.97**]. Table 8 compares OLS, ridge and PCR.

Table 8: 10-fold CV RMSE of OLS, ridge and PCR (`w4_cv_comparison.csv`).

Method	CV RMSE
OLS (full)	[**9.027**]
Ridge ($\lambda_{\min}$)	[**8.329**]
PCR ($k =$ [**9**])	[**8.416**]

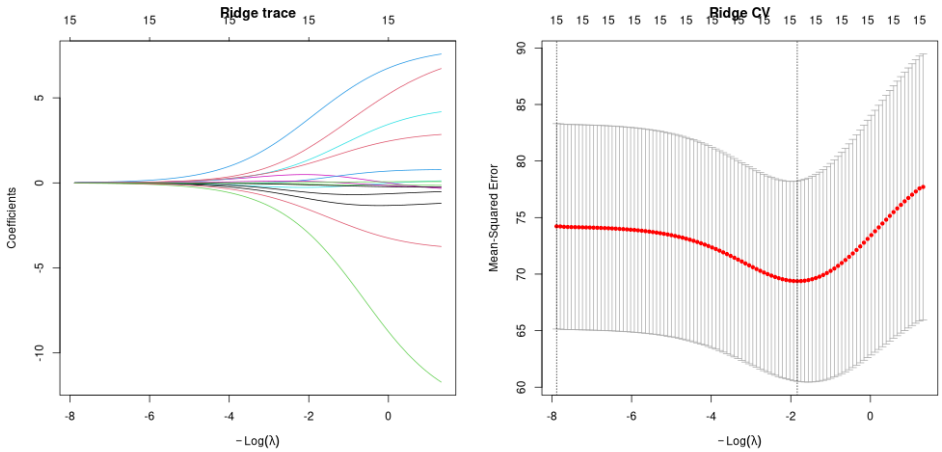


Figure 7: Ridge coefficient paths and cross-validation curve.

4.5 Block tests and curvature

Table 9: Block F-tests (`w4_block_F_tests.csv`).

Block	F	df	p	Holm p
Prior head-coach record	[0.93]	[(5, 68)]	[0.47]	[0.93]
Coordinator background	[2.12]	[(3, 68)]	[0.11]	[0.42]
Ties and pedigree	[0.48]	[(4, 68)]	[0.75]	[0.93]
Hiring context	[1.80]	[(3, 68)]	[0.16]	[0.47]

Curvature tests: [**None of the squared terms for continuous predictors were statistically significant ($p > 0.05$).**].

4.6 Model selection

Best-subsets search suggested [6] predictors under adjusted R^2, [4] under C_p and [2] under BIC. The stepwise formulas were:

- AIC: $\mathrm{sp}_r el_t eam\ PreviousOC + team_s p_r ecent + WonCCasHCany + AgeAtHiringMinus50$
- BIC: $\mathrm{sp}_r el_t eam\ PreviousOC + team_s p_r ecent$

Table 10: Candidate model comparison (`model_comparison.csv`).

Model	p	R^2	Adj. R^2	AIC	BIC	CV RMSE	CV R^2
Null	[1]	[0.00]	[0.00]	[599.62]	[604.49]	[8.62]	[-0.06]
`PreviousOC` only	[2]	[0.10]	[0.09]	[592.73]	[600.03]	[8.19]	[0.05]
BIC-selected	[3]	[0.15]	[0.13]	[589.55]	[599.27]	[8.08]	[0.07]
Confirmatory (full)	[16]	[0.25]	[0.08]	[605.77]	[647.09]	[9.03]	[-0.16]

LASSO and elastic net. The LASSO retained [5] at $\lambda_{\min}$ and [0] at λ_{1se}.

Bootstrap stability. Table 11 gives the proportion of 300 bootstrap samples in which each term was selected. The optimism-corrected RMSE of the BIC model is [8.35], against an apparent RMSE of [7.7].

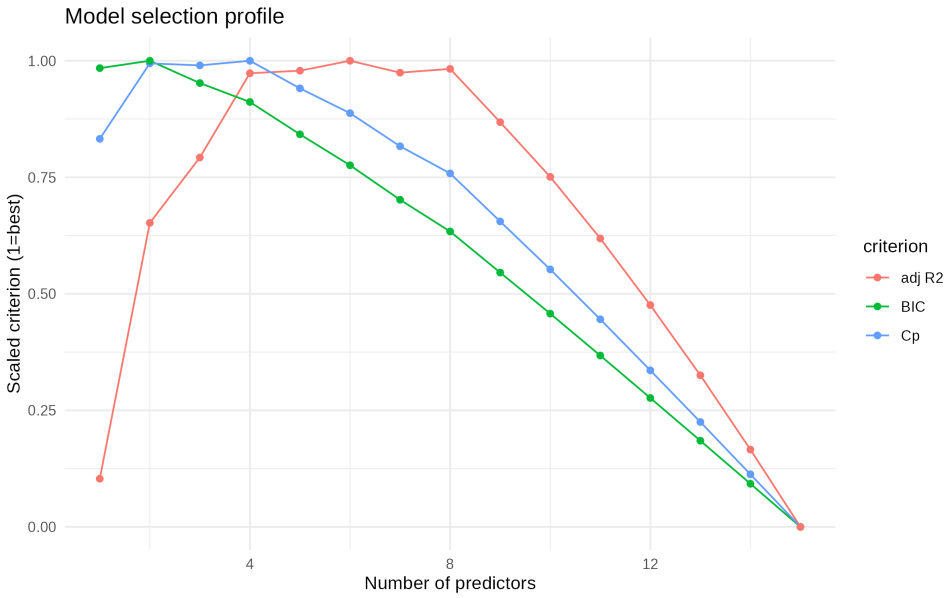


Figure 8: Adjusted R^2, C_p and BIC against subset size.

Table 11: Bootstrap inclusion frequency (`w5_bootstrap_selection_freq.csv`).

Term	Inclusion frequency
`PreviousOC`	[**0.77**]
`team\_sp\_recent`	[**0.48**]
`WonCCasHCany`	[**0.28**]
`PlayedNFL`	[**0.17**]
`AllPriorWinPct`	[**0.13**]
`AgeAtHiringMinus50`	[**0.13**]
`PrevHCAnyLevel`	[**0.11**]
`PromoteWithin`	[**0.10**]
`PreviousWinPct`	[**0.08**]
`ConnectionToHiringSchool`	[**0.08**]
`NFLCoachingExperience`	[**0.06**]
`Conf.LeavingP4`	[**0.05**]
`SideOfBall`	[**0.05**]
`PreviousDC`	[**0.04**]
`PrevHCAnyLevel:sp\_prev\_rel`	[**0.01**]

4.7 Hypothesis tests on prior win percentage

Because `AllPriorWinPct` and `PreviousWinPct` overlap, we examine the effect of prior winning in three ways (Table 12): the coefficient in the full model, the coefficient after dropping `PreviousWinPct`, and the combined effect $\beta_{\text{AllPrior}} + \beta_{\text{Previous}}$ with variance $\text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\beta}_2) + 2\,\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)$.

Table 12: Sensitivity of the prior-win-percentage effect (`h1_sensitivity.csv`).

Specification	Estimate	SE	95% CI	p	Holm p
`AllPriorWinPct` (full model)	[**7.72**]	[**12.33**]	[[**-16.89, 32.33**]]	[**0.53**]	[**1.00**]
`AllPriorWinPct` (`PreviousWinPct` dropped)	[**1.06**]	[**10.18**]	[[**-19.24, 21.36**]]	[**0.92**]	[**1.00**]
Net effect	[**-5.77**]	[**12.43**]	[[**-30.58, 19.04**]]	[**0.64**]	[**1.00**]

4.8 Simultaneous inference

Multipliers: pointwise t = [**1.990**], Bonferroni t = [**2.445**], Scheffé = [**2.855**]. Table 13 shows how the conclusions change as the intervals become more conservative.

Table 13: Coefficients of the BIC model with adjusted inference (`w5_simultaneous_coef_intervals.csv`).

Term	Estimate	Pointwise CI	Bonferroni CI	Raw p	Holm p
`(Intercept)`	[**-1.44**]	[[**-3.76, 0.88**]]	[[**-4.29, 1.41**]]	[**0.22**]	[**0.22**]
`PreviousOC`	[**8.52**]	[[**3.58, 13.45**]]	[[**2.45, 14.58**]]	[**0.001**]	[**0.003**]
`team\_sp\_recent`	[**-0.22**]	[[**-0.42, -0.03**]]	[[**-0.46, 0.02**]]	[**0.026**]	[**0.05**]

After Holm adjustment, `PreviousOC` remains significant ($p = 0.003$), whereas `team_sp_recent` is borderline (Holm $p = 0.052$) and its Bonferroni interval includes zero. Table 14 compares the simultaneous bands for the mean response over all 84 observed covariate patterns.

Table 14: Bands for the mean response (`w5_mean_response_bands.csv`).

Method	Multiplier	Mean half-width
Pointwise t	1.990	2.75
Bonferroni (n points)	3.574	4.95
Working–Hotelling	2.855	3.95

The Working–Hotelling band is narrower than the Bonferroni band, so it is the better choice for simultaneous inference on the whole regression surface.
Conference comparisons (Tukey HSD): [**Tukey's HSD test across Power-4 conferences indicated no statistically significant pairwise differences in coaching impact ($p > 0.05$ for all comparisons), confirming that conference membership does not explain variation in coach performance.**].

4.9 Mixed-effects models

For the school random intercept, $\hat{\sigma}_u^2$ = [**2.521**], $\hat{\sigma}^2$ = [**59.187**] and ICC = [**0.041**]. The boundary-corrected likelihood-ratio test gave χ^2 = [**0.02**], p = [**0.45**]. [**School-level clustering is negligible confirming that a school random intercept is unnecessary and fixed-effects OLS is valid**] Results for conference are in `w5_mixed_summary.csv`.

4.10 Logistic regression for four-year retention

Of the [**84**] coaches, [**66.7**]% were retained for four seasons. The global likelihood-ratio test gave χ^2 = [**22.32**] on [**15**] df (p = [**0.0997**]). McFadden R^2 = [**0.209**] and Nagelkerke R^2 = [**0.324**]. The Hosmer–Lemeshow test gave p = [**0.727**].

Table 15: Odds ratios (`w5_logit_odds_ratios.csv`).

Term	OR	Lower	Upper	p
(Intercept)	[1.20]	[0.01]	[262.71]	[0.95]
PrevHCAnyLevel	[3.07]	[0.31]	[39.06]	[0.35]
AllPriorWinPct	[0.15]	[0.00]	[212.33]	[0.62]
PreviousWinPct	[5.48]	[0.00]	[45237.54]	[0.70]
PreviousOC	[3.19]	[0.32]	[45.82]	[0.35]
PreviousDC	[0.70]	[0.03]	[15.14]	[0.82]
SideOfBallO	[0.08]	[0.00]	[0.75]	[0.05]
ConnectionToHiringSchool	[7.81]	[1.62]	[53.72]	[0.02]
PlayedNFL	[0.71]	[0.10]	[5.68]	[0.73]
AgeAtHiringMinus50	[0.88]	[0.78]	[0.98]	[0.02]
NFLCoachingExperience	[1.37]	[0.35]	[5.84]	[0.65]
WonCCasHCany	[8.66]	[1.40]	[72.28]	[0.03]
Conf.LeavingP4	[0.93]	[0.18]	[4.82]	[0.93]
team_sp_recent	[0.96]	[0.88]	[1.03]	[0.24]
PromoteWithin	[0.29]	[0.03]	[2.62]	[0.28]
PrevHCAnyLevel:sp_prev_rel	[1.03]	[0.93]	[1.15]	[0.54]

Table 16: Discrimination and out-of-sample performance of the logistic model.

Measure	Value
Apparent AUC	[0.781]
10-fold CV AUC	[0.548]
10-fold CV misclassification	[0.452]
Majority-class baseline error	[0.333]
Youden threshold	[0.642]
LASSO-logistic CV AUC	[n/a (null model selected)]

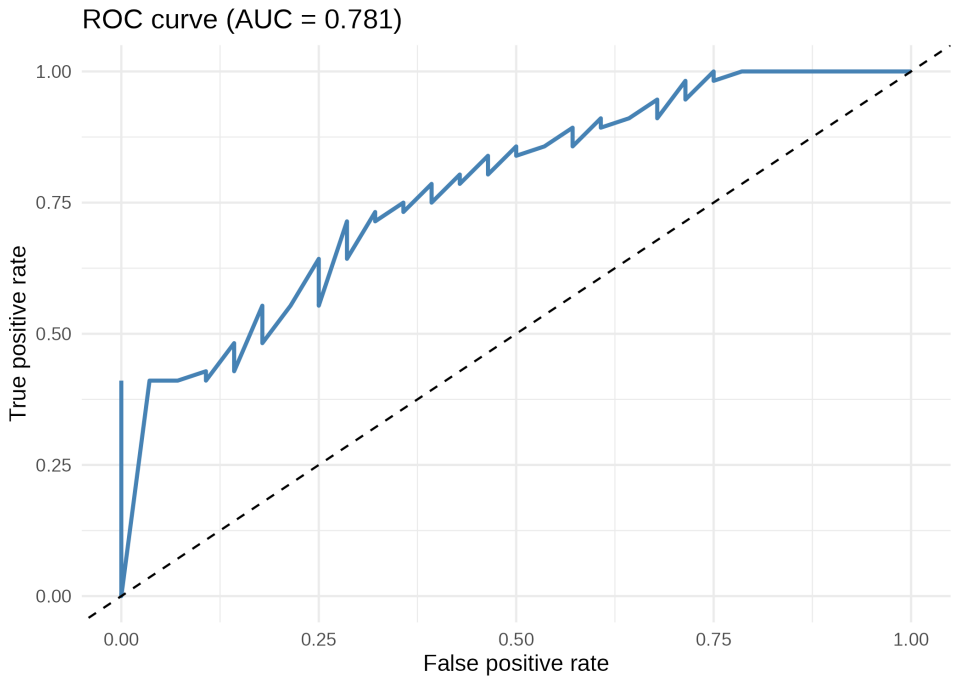


Figure 9: ROC curve of the logistic model.

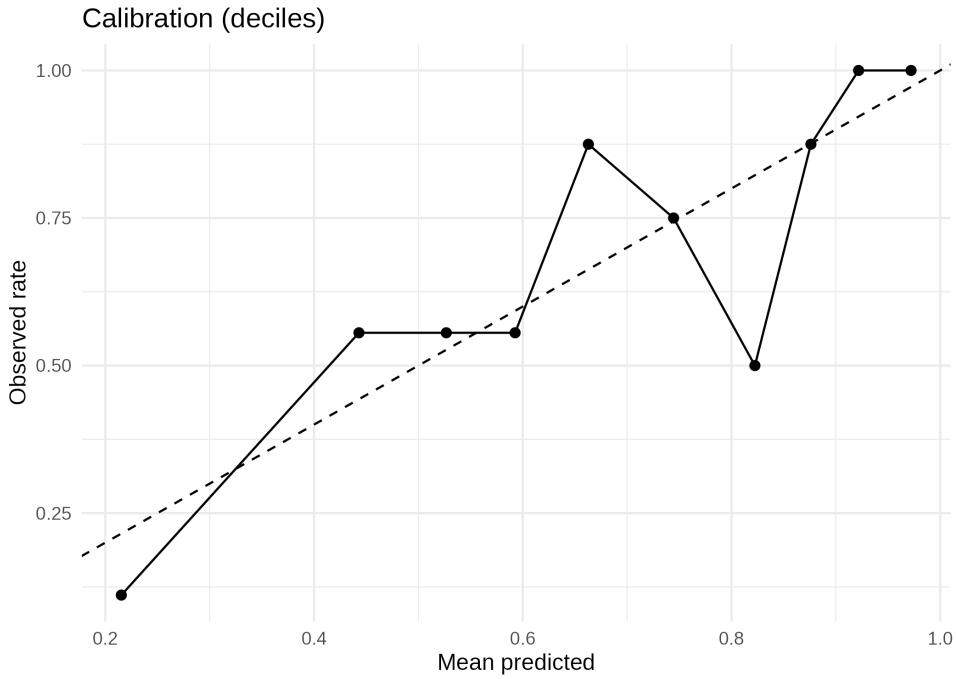


Figure 10: Calibration of predicted retention probabilities by decile.

4.11 Jackknife and bootstrap for `AllPriorWinPct`

Table 17: Resampling inference for the `AllPriorWinPct` coefficient.

Method	Estimate / bias / SE	95% interval
OLS (t-interval)	[7.720, SE 12.332]	[(-16.888, 32.328)]
Jackknife	bias [0.503], SE [13.293]	–
Bootstrap normal	[7.720, bias 0.835, SE 13.486]	[(-19.547, 33.316)]
Bootstrap percentile	[7.720 (median 8.043)]	[(-18.334, 34.679)]

4.12 Survival analysis

The survival sample has $n =$ [120] coaches with [39] events and [81] censored observations. The log-rank test comparing coaches with and without prior head-coaching experience gave $\chi^2 =$ [1.6], $p =$ [0.21].

Table 18: Cox proportional-hazards model (`cox_coefficients.csv`).

Term	Hazard ratio	95% CI	p
`AllPriorWinPct`	[1.75]	[[0.19, 16.26]]	[0.62]
`PrevHCAnyLevel`	[1.22]	[[0.53, 2.77]]	[0.64]
`AgeAtHiringMinus50`	[1.03]	[[0.98, 1.09]]	[0.22]

The proportional-hazards assumption test gave global $p =$ [0.139]. The sensitivity analysis excluding the three flagged coaches (Table 19) [did not] change the conclusions.

Table 19: Survival sensitivity analysis (`survival_sensitivity.csv`).

Version	n	Events	Log-rank p	HR (AllPrior)	HR (PrevHC)
Primary	[120]	[39]	[0.21]	[1.75]	[1.22]
Excluding 3 flagged	[117]	[37]	[0.23]	[1.72]	[1.18]

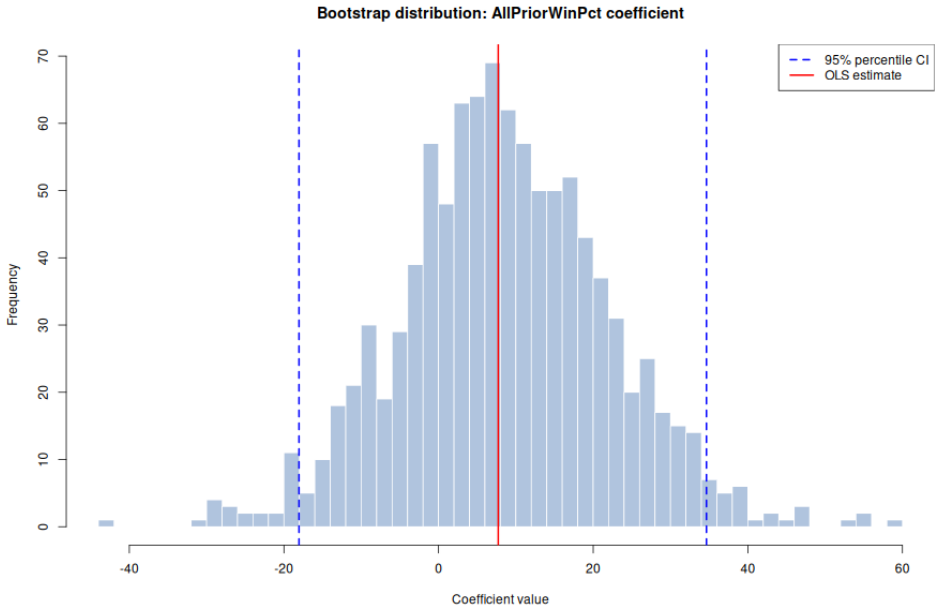


Figure 11: Bootstrap distribution of the `AllPriorWinPct` coefficient.

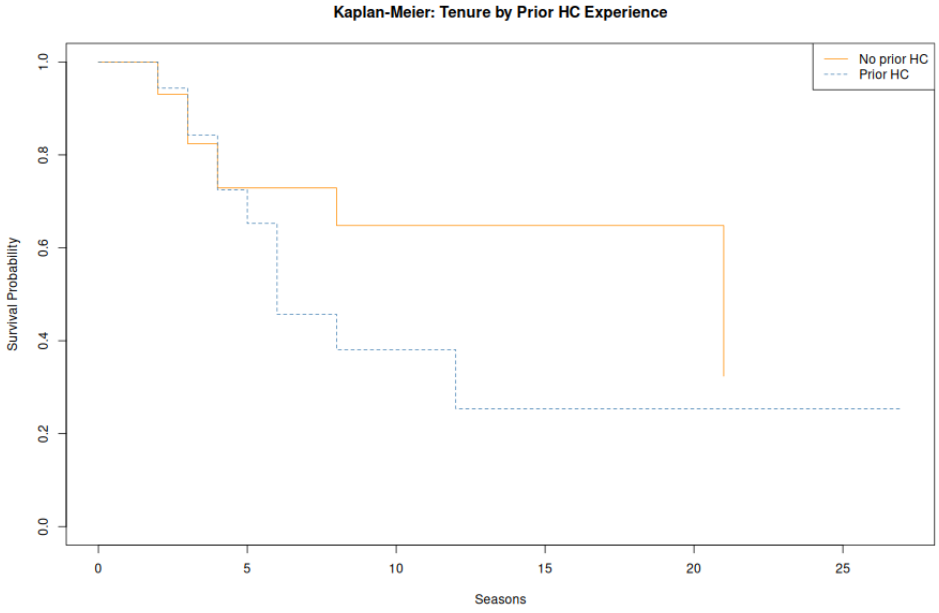


Figure 12: Kaplan–Meier curves of tenure by prior head-coaching experience.

4.13 Holdout validation (2022 onward)

The holdout set contains [42] hires and the training set [75].

Table 20: Holdout performance (`holdout_performance.csv`).

Measure	BIC model	Full model
Training n	[75]	[75]
Holdout n evaluated	[42]	[42]
95% PI coverage	[0.95]	[0.95]
Coverage 95% CI	[[0.84, 0.99]]	[[0.84, 0.99]]
RMSE	[9.00]	[9.16]
MAE	[6.93]	[7.10]

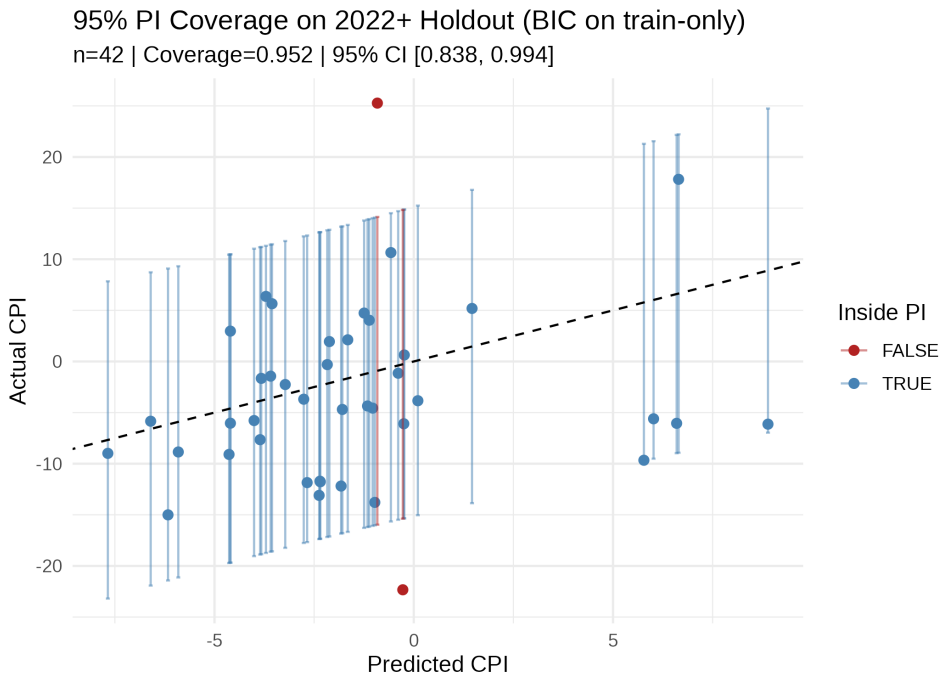


Figure 13: Actual versus predicted outcome with 95% prediction intervals on the 2022+ holdout.

5 Decision Rule: The Coach Hire Score

To make the model usable we built a simple scoring tool. For a candidate with pre-hire characteristics x_0 the model gives a predicted impact $\hat{y}_0$ and a 95% prediction interval. The **Coach Hire Score** places $\hat{y}_0$ within the distribution of fitted values in the training data:

$$\text{Score}(x_0) = 100 \times \frac{\#\{i : \hat{y}_i \le \hat{y}_0\}}{n},$$

so a score of 80 means the candidate is predicted to do better than 80% of past hires.

Three illustrative candidates were scored: **A** an experienced Power-4 head coach moving from a strong programme, **B** a young first-time head coach going to a weak programme, and **C** an experienced offensive coordinator with ties to the hiring school going to a mid-level programme.

Table 21: Coach Hire Score for the example candidates (`coach_hire_score_examples.csv`).

Candidate	Model	Predicted	Lower	Upper	Score
A (established HC)	BIC	[-3.22]	[-18.95]	[12.51]	[45.2]
A (established HC)	Full	[-0.99]	[-18.38]	[16.40]	[63.1]
B (first-time HC)	BIC	[6.63]	[-9.73]	[23.00]	[97.6]
B (first-time HC)	Full	[6.16]	[-12.38]	[24.71]	[95.2]
C (experienced OC)	BIC	[6.41]	[-9.93]	[22.75]	[96.4]
C (experienced OC)	Full	[3.97]	[-15.71]	[23.65]	[88.1]

Because the prediction intervals are wide relative to the differences between candidates, the score should be used to *rank and support* decisions, not to make them automatically.

6 Discussion and Limitations

- **Small sample.** With roughly [84] observations and about 15 predictors, estimates are noisy and prone to overfitting. This motivated cross-validation, penalised methods and the bootstrap.
- **Single imputation.** Replacing missing values by 0.50 or the sample mean ignores imputation uncertainty. Standard errors may be slightly too small.
- **Post-selection inference.** Confidence intervals and p-values from the BIC model do not account for the fact that the model was chosen from the data. We therefore also report the full confirmatory model and the optimism-corrected error.
- **Collinearity.** `AllPriorWinPct` and `PreviousWinPct` carry overlapping information, so their individual coefficients are unstable although their combined effect is better determined.
- **Measurement and data quality.** Some employment fields were inconsistent and a few records were excluded or flagged. SP+ is itself a model-based rating.
- **Holdout size.** The 2022+ holdout is small and recent, so the coverage estimate has a wide confidence interval and some coaches had few completed seasons.
- **Observational data.** Associations are not causal. Better candidates may also be hired by better-resourced schools.

7 Conclusions

1. **Predictors.** `PreviousOC` is the strongest predictor: having been an offensive coordinator is associated with about +8.5 SP+ points, it was selected in 77% of bootstrap samples, and it remains significant after Holm adjustment ($p = 0.003$). `team_sp_recent` has a negative association (-0.22 per SP+ point), was selected in 48% of bootstrap samples, and is borderline after Holm adjustment ($p = 0.052$). Prior win percentage (`AllPriorWinPct`) is not significant under any method.
2. **Model validity.** The Breusch–Pagan test ($p = 0.398$), White's test ($p = 0.833$) and the Durbin–Watson test ($p = 0.736$) show no violations of constant variance or independence. The Box–Cox estimate $\hat{\lambda} = 0.9$ indicates that no transformation was needed. Collinearity is mild (largest VIF = 5.5, condition number = 6.7).
3. **Predictive performance.** The BIC-selected model has a cross-validated R^2 of 0.07 (null model: -0.06) and a CV RMSE of 8.08 against 8.62 for the null model, a small improvement. On the 2022+ holdout the RMSE is 9.00, which is not lower than the standard deviation of the outcome (8.44), so the out-of-sample gain over simply predicting the mean is negligible. The 95% prediction-interval coverage (0.95) is largely due to very wide intervals (about ± 16 SP+ points).
4. **Clustering.** School and conference effects are negligible: the intraclass correlation is 0.04 for school and 0.005 for conference, and the likelihood-ratio tests are not significant.
5. **Retention and tenure.** The logistic model for four-year retention does not generalise: its cross-validated AUC is 0.548 (apparent AUC 0.781), and both BIC selection and the LASSO choose the intercept-only model. In the Cox model, none of the hazard ratios is significant (for example, `AllPriorWinPct` has HR = 1.75, 95% CI 0.19–16.26).
6. **Practical use.** The Coach Hire Score gives a transparent way to compare candidates, but its wide prediction intervals mean it should supplement, not replace, expert judgement.

Future work: multiple imputation for missing values, more seasons of data, hierarchical models with school-specific slopes, and non-linear methods such as random forests.